

Allan Standard Deviation to Gauss-Markov State-Space Model With Itô Analysis

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Abstract

This tutorial article motivates and describes the process for extracting a first-order, Gauss-Markov, state-space model corresponding to an Allan Standard Deviation (ASD) plot over a range of delays. The purpose of this model is to be used in a state estimator to represent the stochastic errors on the IMU measurements. One means to test the sufficiency of the state-space model is that the data output by a simulation of the state-space model should reproduce the ASD plot.

The methods in this document apply to both gyros and accelerometers. To be generic, the article will use the terminology of phase and frequency. The term *frequency* refers to either the IMU angular rate or specific force measurement. The term *phase* refers to the integral of frequency, which for an IMU refers to angle or velocity. This nomenclature as summarized in Table 1 is standard.

Given values for the parameters: N , B , T_p , and F_s , the script `ASD_to_GaussMarkovFirstOrder.py` available at <https://github.com/jaffarrell/AV-Matlab-SW.git> implements all the computations that are presented herein in a blue font.

Appendices are included: Appendix A analyzes noise properties rigorously using Ito calculus. Appendix B gives examples working through the units for gyro and accelerometer bias driving noise power spectral densities. Appendix C derives an approximation that is commonly used. Additional discussion, such as methods to compute the Allan variance, can be found in [1].

1 Introduction

For Bayesian optimal estimation in applications involving IMU's, one of the most important items is the IMU error model. This model communicates to the estimator the magnitude and temporal correlation structure of the IMU errors. This is particularly important for optimally combining IMU information with information extracted from other sensors, e.g.: vision, GNSS, lidar, radar.

For the benefit of the optimal estimation community, particularly users of GTSAM, we are writing this document that describes the method for converting IMU datasheet specifications to state-space models for IMU errors in a form suitable for optimal estimation algorithms.

Symbol	Clock	Gyro	Accelerometer
$\Omega(t)$	frequency	angular rate	specific force
$\theta(t)$	phase	angle	velocity

Table 1: Equivalent terminology for Allan Variance Symbols

Any real physical system is impossible to model perfectly. Often such systems can be modeled sufficiently for a well-defined purpose by a linear system.

The purpose of the IMU error model is to communicate to an optimal state estimation routine, the accuracy of the IMU measurements so that it can properly incorporate them into its data processing. One axis of the accelerometer (i.e., specific force) or gyro measurement in IMU frame can be modeled as

$$y(t) = s(t) + z(t) + d(t) \quad (1)$$

where $s(t)$ represents the signal (i.e., specific force or angular rate) the instrument is designed to measure, $z(t)$ represents the stochastic errors, and $d(t)$ represents non-stochastic errors (e.g., scale factor, non-orthogonality, saturation). This document is only concerned with the stochastic errors: $z(t)$. The non-stochastic errors are addressed by other methods.

1.1 ASD Parameters

Manufacturers often follow the IEEE standard 952-2020 [2] and specify their IMU's typical stochastic errors through either an ASD plot or selected parameters extracted from it. Table 2 summarizes this nomenclature. Each row corresponds to one type of independent noise. The first and last columns list the noise names. The three bold names are typically the dominant noise types in IMU applications. The two middle columns summarize typical symbols used in the literature. For sensors that do not come with extensive quantified parameters, one could measure the ASD using a setup similar to those in [2], and use toolboxes such as [3] to compute the ASD.

1.2 Stochastic Error: PSD and ASD

The Power Spectral Density (PSD) and ASD of the stochastic error $z(t)$ are directly related to each other. Each can be written as an infinite sum of terms involving the parameters summarized in Table 2. These parameters can be directly read from either the ASD or PSD curve. The relations between the most important terms in each series are derived in [2].

1.2.1 ASD plots

The ASD can be represented as an infinite sum of terms [2]:

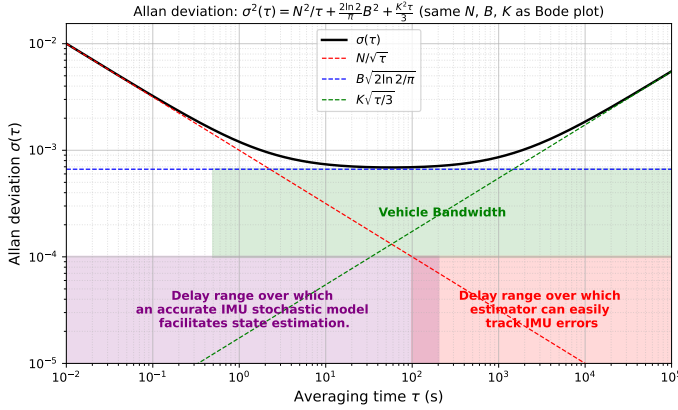
$$\sigma^2(\tau) = \dots + \sigma_{z_N}^2 + \sigma_{z_B}^2 + \sigma_{z_K}^2 + \dots, \quad (2)$$

$$\approx \dots + \frac{N^2}{\tau} + \frac{2B^2 \ln 2}{\pi} + \frac{K^2 \tau}{3} + \dots, \quad (3)$$

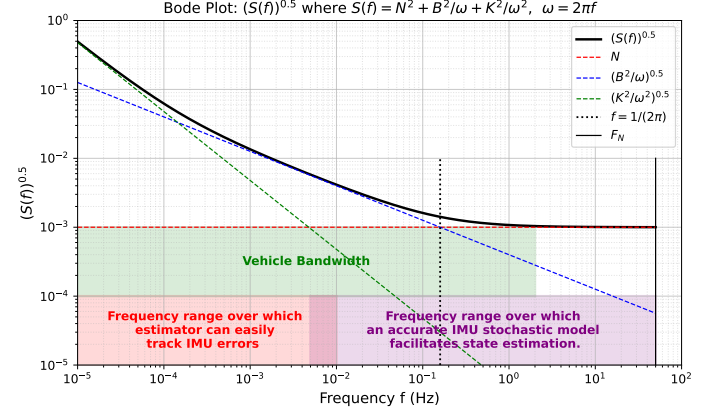
An example ASD plot is shown in Fig. 1a. The method of reading the ASD parameters from this plot is described in Section 3.

Nomenclature	Other name		
	Generic	IEEE 952	
White Angle	N_θ^2	Φ^2	Angle measurement noise
Angle quantization	—	Q^2	White angle energy spectrum
Flicker angle	$N_{F\theta}^2$	—	Pink angle noise
Angle random walk	N_Ω^2	N^2	Red angle noise
Rate quantization	—	—	Discrete white rate noise
Bias instability	$N_{F\Omega}^2$	B^2	Pink rate noise
Markov rate	$N_{c\Omega}^2$	q^2	Correlated drift rate
Rate random walk	N_Ω^2	K^2	Red rate noise
Ramp instability	$N_{F\Omega}^2$	R^2	Pink angular accel. noise
Random bias	$\Omega(0)$	B_0^2	Bias/fixed drift
Random ramp	$\dot{\Omega}(0)$	R_0^2	Rate ramp
Periodic rate	Ω_0	—	Harmonic

Table 2: Stochastic model contributors and their gyro nomenclature [2]



(a) Allan Standard Deviation (black) with the N , B , and K asymptotes drawn in red, blue, and green.



(b) Root Power Spectral Density (black) with the N , B , and K asymptotes drawn in red, blue, and green.

Figure 1: ASD and PSD plots that also depict the vehicle bandwidth and the range of delays or frequencies over which an accurate IMU stochastic error model will facilitate the discrimination between random measurement noise and correlate (i.e., bias) errors. Both figures are drawn for the parameter values: $F_s = 100$ Hz, $N = 0.001^\circ/\sqrt{s}$, $B = 0.001^\circ/s$, $K = 0.00003^\circ/s/\sqrt{s}$.

1.2.2 PSD

The PSD can be represented as an infinite sum of terms [2]:

$$S_z(f) = \dots + N^2 + \frac{B^2}{2\pi f} + \frac{K^2}{(2\pi f)^2} + \dots, \quad (4)$$

where f denotes frequency in Hz. Figure 1b shows the PSD corresponding to the ASD plot in Fig. 1a.

Each even term in this series (e.g., N^2 and $K^2/(2\pi f)^2$) can be perfectly represented by a finite dimensional continuous-time linear system. For odd terms (e.g., $B^2/(2\pi f)$), there is no finite dimensional linear system to represent any one of them [1].

Due to (1) our inability to model the odd terms and (2) there being an infinite number of even terms, there does not exist a perfect finite dimensional linear system model for the IMU stochastic error. Therefore, an approximate model is required.

1.2.3 Model: Which Terms are Important

Fortunately, due to the nature of the problem and high quality design by the manufacturer, only a few terms of the infinite series are important when determining the approximate model.

Figure 1 is included to help illustrate the following points:

- The navigation engineer, understanding aliasing and Nyquist frequency ideas will ensure that the sample frequency F_s of the IMU is much larger than the bandwidth (F_B) of the vehicle on which it will be mounted (i.e., $F_s \gg F_B$). The ASD plot has its smallest delay at the sampling period ($T_s = 1/F_s$). The PSD plot has its highest frequency at the Nyquist frequency $F_N = F_s/2$. All important angular rate and acceleration signal information is within the vehicle bandwidth, at the much lower frequencies (much higher delays) indicated by the green shaded region.
- The manufacturer will ensure that the measurement quantization error (i.e., Q) is much smaller than the instrument measurement noise (i.e., N). Therefore, for frequencies $F_B < f \leq F_N$ (i.e., delays $2T_s \leq \tau < 1/F_B$) the phase white random walk noise (N) will dominate.
- At mid-to-lower frequencies (i.e., higher delays) time-correlated errors become important. For these intermediate frequency ranges, the bias instability (parameter B) and bias random walk (parameter K) are dominant. The designer of the

navigation system can include states in the estimator to model these time-correlated states.

- Because the navigation system can accurately track the very low-frequency (i.e., very long-delay) portion of these stochastic errors, this portion of the PSD (or ASD) curve, indicated by the red shaded region, is less important for the error model.

Over the intermediate to high frequencies (i.e., intermediate to low delays) indicated in purple the N , B , and K parameters typically dominate. A state-space error model that accurately represents the PSD and ASD curves over this range of frequencies (or delays) facilitates the job of the estimation system in distinguishing between white and time-correlated errors. The exact boundary between the red and purple regions is determined by many design factors: IMU specifications, desired performance, and computational budget.

1.3 Finite Dimension, Gauss-Markov Model

Section 1.2 explained why, typically, only three parameters (N , B , and K) are necessary to model the stochastic errors over the frequency (delay) range of interest. However, due to the B term, which appears as an odd-term in the PSD series of eqn. (1.2), these three terms theoretically still require an infinite order linear system to model them exactly.

More practically, the two N and K terms can be perfectly modeled by a first-order Gauss-Markov model. This N and K only model, may model the stochastic error well at short and long delays, but not adequately model it in the mid-range of delays corresponding to the flat portion of the ASD curve. That flat portion is often more important to performance than the long delay portion. The three N , B , and K terms can also often be adequately modeled, over the frequency (delay) range of interest, by a first-order Gauss-Markov model.

1.4 Why First-Order?

The order of the stochastic error model is important because the computational load of the matrix operations. For example, in a Kalman filter that load is $O(n^3)$, where n is the order of the state vector. That state vector includes the vehicle state error ($n_v = 9$), the number of states ($6n_d$) used to model $d(t)$, and the number of states ($6n_z$) used to implement the model of $z(t)$. The factor of 6 comes from the fact that the IMU has 6 measurements. Therefore the computational load is on the order of $O((n_v + 6(n_d + n_z))^3)$.

The purpose of the Gauss-Markov model is to communicate to the estimation algorithm the frequency distribution (i.e., time-correlation structure) of the stochastic errors. If the ASD plot over the important range of delays (or the PSD plot over the important range of frequencies) can be adequately represented by a first-order model, there is a high computational penalty for implementing a higher-order model.

2 First-Order, Gauss-Markov Models

This section summarizes a few facts about first-order Gauss-Markov state-space models.

2.1 Continuous-Time

The continuous-time model has the form¹

$$z(t) = b(t) + n(t), \quad \text{where } n(t) \sim \mathcal{N}(0, S_n) \quad (5)$$

$$\frac{db(t)}{dt} = -\mu b(t) + \zeta(t), \quad \text{where } \zeta(t) \sim \mathcal{N}(0, S_\zeta), \quad (6)$$

and the decay rate $\mu \geq 0$. The symbols $n(t)$ and $\zeta(t)$ represent independent, continuous-time, Gaussian, white noise processes with power spectral densities of S_n and S_ζ , respectively. For the discussion herein, these PSD's are assumed to be constant.

This model is discussed in Example 4.19 on p. 140 in [4], which shows that the time evolution of the $P(t) = \text{cov}(b(t))$ is described by

$$\dot{P}(t) = -2\mu P(t) + S_\zeta. \quad (7)$$

For $\mu > 0$, as t increases towards infinity, $\dot{P}(t)$ approaches 0, and $P(t)$ approaches its steady-state value (see derivation of eqn. (37))

$$P_\infty = \frac{S_\zeta}{2\mu}. \quad (8)$$

The model parameters (i.e. μ , S_n , and S_ζ) can be estimated from the ASD plot, as will be discussed in Section 3.

Remark 2.1: If $\mu = 0$, the bias is modeled as a pure random walk. This case is analyzed in Appendix A.4, which shows that the system does not have steady-state covariance as $t \rightarrow \infty$; instead, $P(t)$ grows to infinity, which is not a realistic covariance for an IMU bias. $\triangle$

2.2 Discrete-time

The discrete-time model has the form

$$z_k = b_k + n_k, \quad \text{where } n_k \sim \mathcal{N}(0, \sigma_n^2) \quad (9)$$

$$b_{k+1} = \phi b_k + \zeta_k, \quad \text{where } \zeta_k \sim \mathcal{N}(0, \sigma_\zeta^2), \quad (10)$$

where $|\phi| \leq 1$. The subscript k on any variable denotes the value of the variable at the discrete instant of time $t_k = kT_s$, where $T_s > 0$ is the IMU sampling rate. For example, $z_k = z(kT_s)$. The symbols n_k and ζ_k represent independent, Gaussian, discrete-time, white noise processes with covariances of σ_n^2 and σ_ζ^2 , respectively. For the discussion herein, these covariances are assumed to be constant.

The time evolution of the $P_k = \text{cov}(b_k)$ is described by

$$P_{k+1} = \phi^2 P_k + \sigma_\zeta^2. \quad (11)$$

For $|\phi| < 1$, as $k \rightarrow \infty$, a steady-state is reached wherein $P_{k+1} = P_k$; therefore, the steady-state covariance satisfies

$$P_\infty = \frac{\sigma_\zeta^2}{1 - \phi^2}. \quad (12)$$

The parameters (i.e., ϕ , σ_n^2 , and σ_ζ^2) are discussed in Section 2.3.

Remark 2.2: If $\phi = 1$ (i.e., the bias is modeled as a discrete-time random walk), no steady-state exists, and $P_k \rightarrow \infty$ as $k \rightarrow \infty$. $\triangle$

¹A more rigorous analysis using Ito calculus is presented in Appendix A.5.

2.3 Parameter Choice for Equivalent Models

The continuous and discrete-time models will represent equivalent models, in the sense that their noise statistics are equal at the times $t_k = k T_s$, if the discrete-time model parameters are selected as

$$\phi = \exp(-\mu T_s), \quad \sigma_n^2 = S_n F_s, \quad \text{and} \quad \sigma_\zeta^2 = \frac{1-\phi^2}{2\mu} S_\zeta. \quad (13)$$

Appendix B presents a unit analysis to check the final equation.

2.4 Modeling Approach Summary

The continuous-time model parameters are easily read from the ASD plot. Translation to an equivalent discrete-time model is useful for a few reasons:

- Discrete-time estimators are easier to implement and require discrete-time models.
- The discrete-time model can easily be simulated. If it is correct, then the ASD plot made from its simulated data should match the ASD plot of the instrument of interest.

The next section describes the process for extracting the model parameters from the ASD plot.

3 ASD Graph to State-Space Model

The goals of this section are: to show how to extract parameters N , B , and T_p from an ASD plot; to show how to use those parameters to define a first-order, continuous-time, state-space model; to demonstrate the use of the equations in (13) to find an equivalent first-order, discrete-time state-space model; and, to demonstrate that the discrete-time equivalent model will generate data that reproduces the ASD graph.

Samples along the ASD curve for one axis of an accelerometer are shown as blue dots in Figs. 2a - 2c. The tangent lines discussed in the following are also shown in Fig. 1a.

1. Find the portion of the ASD graph that has a slope of -1/2. Draw a tangent line, and define phase random walk parameter N as the value of this ASD tangent line where it crosses the vertical line at $\tau = 1$.

In Fig. 2a, the slanted orange line with slope -1/2 intersects the vertical line at $\tau = 1$ for $N = 3.3\text{e-}3 \frac{m/s^2}{\sqrt{Hz}}$.

2. The values of the bias instability S_ζ and T_p are selected together.

- The value of the bias instability parameter B is calculated from the value of the ASD plot at which its slope is zero.

For many inexpensive IMU's this flat section may not exist or may be difficult to select due to the error in the computed values the ASD plot. The ASD plot in Fig. 2a is very clean and a reasonable value of the ASD for its flat region is indicated by the dashed green line with the value

$$ASD_B = 8.0 \text{ e-}4 \text{ m/s/s}. \quad (14)$$

Using eqn. (32) in [1],

$$ASD_B = \sqrt{\frac{2\ln(2)B^2}{\pi}} = 0.664 B,$$

the bias instability parameter can be computed as

$$B = 1.505 ASD_B = 1.2 \text{ e-}3 \text{ m/s/s}.$$

This approach to determine B is presented for completeness, but is not explicitly used in the development of the state-space model.

- As discussed in Section 1.2.2, there is no finite-dimensional state space-model that can exactly reproduce the bias instability.

As motivated by Section 1.4, we select the first-order Gauss-Markov model of eqn. (6) to approximate the bias instability portion of the ASD plot. The shape of the ASD plot corresponding to this model is defined by eqn. (35) in [4]:

$$\sigma_{zb}^2(\tau) = \frac{S_\zeta T_B^2}{\tau} \left[1 - \frac{T_B}{2\tau} \left(3 - 4e^{-\tau/T_B} + e^{-2\tau/T_B} \right) \right],$$

which is sketched as black curve with dashed green asymptotes in Fig. 2b. Its asymptotes have slope $\pm 1/2$; therefore, for delays near, but smaller than T_B , the parameters S_ζ and T_B can be selected to produce a flat region with the desired height.

This expression has a peak value at $T_p = 1.89 T_B$ of

$$\sigma_{zb}^2(1.89 T_B) = \left(0.4365 \sqrt{S_\zeta T_B} \right)^2,$$

see eqn. (37) in [1]. Solving for the PSD yields

$$S_\zeta = \frac{ASD_B^2}{(0.4365)^2 T_B}.$$

We chose the peak value to be the ASD_B value from eqn. (14). A bit of trial-and-error shows that choosing the peak to be at $T_p = 100$ seconds (i.e., $T_B = 52.91$), yields the ASD curve for the Gauss-Markov (i.e., bias portion) of the model is depicted as orange dots in Fig. 2b with

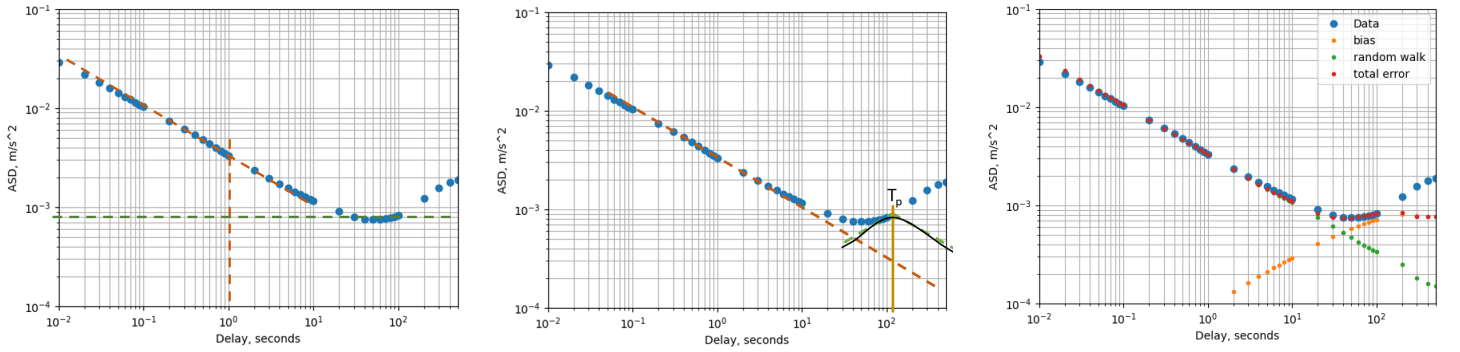
$$S_\zeta = 6.35\text{e-}8 \frac{(m/s^3)^2}{Hz}.$$

The overall model combining the random walk and Gauss-Markov bias model produces the total ASD curve depicted in red in Fig. 2b. It matches the ASD graph for the actual data for $\tau \in [0.01, 100]$ seconds.

To match the ASD curve for $\tau > 100$ seconds would require adding another state. If the designer is interested, they can implement and compare the performance and computational load of the two modeling approaches. Herein, we proceed with the one-state model.

4 ASD Error Modeling

After extracting the Allan variance parameters using the methods of Section 3. You can use them to generate the continuous and discrete time state space model parameters. Section 4.1 summarizes the formulas for defining the state space model parameters. Those formulas and tools for evaluating the resulting state space model are already implemented in Python, as discussed in Section 4.2.



(a) Allan Standard Deviation Plot marked for estimation of N and B .

(b) Allan Standard Deviation Plot marked with approximations for the first-order Gauss-Markov model.

(c) Allan Standard Deviation Plot for data, the first-order Gauss-Markov model, and its random walk and bias components.

Figure 2: ASD plots supporting the discussion of Section 3.

4.1 Analytic Models

The continuous-time phase random walk power spectral density is $S_n = N^2$, from eqn. (22) in [1]. For this example with $N = 3.3e-3 \frac{m/s^2}{\sqrt{Hz}}$, we compute $S_n = 1.09e-5 \frac{(m/s^2)^2}{Hz}$. The values of T_B and S_ζ are selected together as discussed in Item 2 of Section 3. Given these values of T_B and S_ζ , the decay rate and steady-state covariance are:

$$\mu = \frac{1}{T_B} \quad \text{the decay rate in eqn. (6)}$$

$$\bar{P}_\infty = \frac{S_\zeta}{2\mu} \quad \text{steady state covariance of } b(t) \text{ by eqn. (8)}.$$

For the example values herein this produces:

$$T_B = 52.91 \text{ sec} \quad \mu = 1.89e-2 \text{ Hz}$$

$$S_\zeta = 6.35e-8 \frac{(m/s^3)^2}{Hz} \quad \bar{P}_\infty = 1.68e-6 (m/s^2)^2.$$

Units are reviewed in Appendix B. Using eqn. (13) with the parameter values above yields the discrete-time model parameters:

$$\phi = 0.9998, \quad \sigma_n^2 = 1.089e-03 (m/s^2)^2, \quad \sigma_\zeta^2 = 6.3473e-10 (m/s^2)^2.$$

4.2 Python Tools

The python section of the repository at <https://github.com/jaffarrell/AV-Matlab-SW.git> contains python scripts: to convert ASD parameters to both continuous-time and discrete-time models, to simulate the discrete-time model discrete-time model, and to simulate the model to produce sample data, points to the *allantools* repository for python code to compute and graph the Allan standard deviation. It also includes tools to compare the theoretical and sample error covariance as a function of time. Finally, it also includes a Jupyter note book that demonstrates the use of the python script. This section shows example results.

4.2.1 Model Output Results

For the ASD parameters $N = 3.3e-3 \frac{m/s^2}{\sqrt{Hz}}$, $ASD_B = 8.0 e-4 m/s/s$, $T_p = 100$, the Jupyter notebook outputs the following models:

- Continuous-time model is (units depend on context):

$$z(t) = b(t) + n(t),$$

$$db(t)/dt = -1.8900e-02 * b(t) + w(t),$$

where $n(t)$ is white noise with PSD $S_n = 1.089e-05$ and $w(t)$ is bias process noise with PSD $S_w = 6.3485e-08$. The steady-state covariance of the bias process is $P_{b_\infty} = 1.6795e-06$.

- Discrete-time model is (units depend on context):

$$z[k] = b[k] + n[k],$$

$$b[k+1] = 0.9998 * b[k] + w[k],$$

where $n[k]$ is white noise with covariance $Q_n = 1.0890e-03$ (std $3.300e-02$) and $w[k]$ is the bias process noise with covariance $Q_w = 6.3473e-10$ (std $2.5194e-05$). The steady-state covariance of the bias process is $P_{b_{ss,d}} = 1.6795e-06$.

Note that both models have the same steady-state covariance for the bias, as they should.

4.2.2 Data generation for Model Checking

The discrete-time model is easily simulated to yield data sequences that can be used to check the validity of the error model. Two validity checks are considered. First, the ASD computed from the simulated data should match that of the actual data for the range of delays that is desired. Second, the ensemble of trajectories computed by integrating either the real or simulated data should have covariance growth versus time that matches the theoretical prediction of eqn. (11). These methods are illustrated in the next two subsections.

4.2.3 Check: ASD Plot

Fig. 2c shows the ASD graph for the simulated output of the state-space error model as red dots. These closely match the ASD plot for the instrument data (blue) for delay $\tau \in [T_s, 100]$ seconds. The figure also shows the ASD plot for both components b (orange) and n (green) that are summed to create the overall output z .

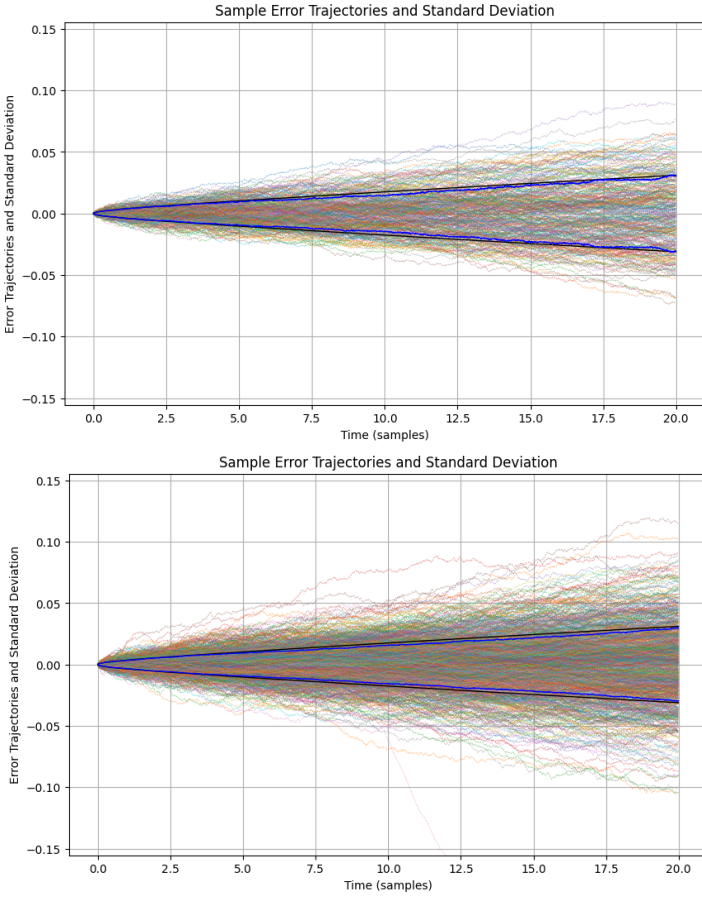


Figure 3: Covariance Check Graphs. Top: One thousand sample error trajectories computed by integrating the output of the state-space error model. Bottom: One thousand sample error trajectories computed by integrating the IMU data after subtracting a prior estimate of the bias. Black solid curve is the theoretical standard deviation from the state-space model. Blue solid curve is the sample standard deviation at each time instant over the sample trajectories.

4.2.4 Check: Covariance

The main purpose of the model is to communicate to the optimal estimation routine the magnitude and correlation structure of the IMU error covariance matrix. The IMU outputs are integrated to produce angles or velocities. Therefore, this section compares the growth of the integrated error in a few ways:

- The error model produces 1000 simulated integrated error trajectories. Each is plotted as a narrow opaque curve in the top portion of Fig. 3. At each time instant the robust standard deviation is computed and plotted as a blue line. The theoretical error covariance is computed using the state-space model and eqn. (11) (i.e., eqn. (4.125) in [4]). This is plotted as the black line. Note that the blue and black lines match closely at all times.
- After removing a constant prior estimate of the bias, the instrument data produces 1000 simulated integrated error trajectories. Each is plotted as a narrow opaque curve in the bottom portion of Fig. 3. At each time instant the robust standard deviation is computed and plotted as a blue line. The theoretical error covariance is computed using the state-space model and eqn. (11) (i.e., eqn. (4.125) in [4]). This is plotted as the

black line. Note that the blue and black lines match closely at all times.

The fact that the theoretical (black) and ensemble (blue) standard deviation curves for both the real and simulated data both match demonstrate that the model can serve its purpose to the estimator. This curve is for a duration of 20 seconds which well within the range of delays over which the model was expected to match the data. As the duration of the simulation approaches and surpasses this range the models ability to predict the integrated error covariance will break down.

5 Using the IMU Error Model in GTSAM

GTSAM represents the measurement likelihood by Gaussian factors in a nonlinear factor graph. Sections 2 and 3 derived the continuous-time and discrete-time IMU error models, and the preceding covariance checks verified that those models reproduce the stochastic behavior over the delay range of interest. This section explains how the same quantities map into GTSAM classes and how the choice of factorization should match the shape of the ASD curve.

5.1 Preintegrated IMU Measurements

For computational efficiency, GTSAM does not usually add one factor per IMU sample. Instead, it accumulates the samples between two keyframes into a `PreintegratedImuMeasurements` object and then uses that object inside either an `ImuFactor` or an `ImuFactor2` [5, 6]. The PIM stores the preintegrated change in rotation, velocity, and position, the accumulated integration time, and the associated error covariance. During this process the integration is linearized about a fixed bias estimate, denoted `biasHat` in GTSAM, and the resulting factor later applies first-order bias corrections about that linearization point [6].

When bias evolution is modeled inside the preintegration object, GTSAM instead uses `PreintegratedCombinedMeasurements` together with a `CombinedImuFactor`. In that case the stored covariance is larger because it also includes the correlations with the bias states [6].

5.2 Measurement Noise Mapping

Section 2.3 showed that the continuous-time white measurement-noise PSD S_n and the discrete-time covariance σ_n^2 are related by eqn. (13). GTSAM's `PreintegrationParams` expects the continuous-time IMU noise power spectral density, not the per-sample covariance already scaled by the sample interval [6]. Therefore the quantity passed to `setGyroscopeCovariance()` or `setAccelerometerCovariance()` should be S_n . Similarly for the bias noise parameters (i.e., `biasAccCovariance` or `biasOmegaCovariance`) are set to S_ζ . Both with units consistent with the discussion in Appendix B. GTSAM then performs the accumulation over each `integrateMeasurement(dt)` call internally.

Consequently, the ASD parameter extraction carried out earlier in the paper maps directly into GTSAM: for each axis, the PSD-style quantities inferred from the ASD curve define the entries of the continuous-time covariance matrices supplied to

PreintegrationParams. These parameters should not be converted by hand into ad hoc sample-to-sample variances before being passed to the PIM.

5.3 Bias Model Choices

GTSAM provides two built-in ways to connect the bias states across keyframes. The first is to use an `ImuFactor` or `ImuFactor2` together with a `BetweenFactorConstantBias`. The second is to use a `CombinedImuFactor` together with `PreintegrationCombinedParams` and `PreintegratedCombinedMeasurements`. The official GTSAM documentation shows that these two constructions are mathematically equivalent when they use the same continuous-time bias random-walk PSD [7]. In both default constructions the bias model is Brownian: the discrete covariance accumulated over an interval is proportional to the interval duration, and no steady-state bias covariance exists. The `ImuFactor` (or `ImuFactor2`) allows the bias state transition model to be implemented outside; therefore, the user can choose a between factor implementation of Brownian motion model, the Gauss-Markov model (e.g., eqns. (5) and (6)), or any separate (e.g. user-implemented) factor with dynamics chosen to match the fitted IMU error model.

The built-in bias model of the `CombinedImuFactor` uses $\mu = 0$ or $\phi = 1$, which is the limit of the first-order model of Section 2. It is appropriate when a pure random walk is an adequate approximation over the delay range of interest. It is not, however, the same stochastic assumption as the finite-correlation-time Gauss-Markov approximation advocated earlier in the paper for a pronounced flat bias-instability region.

5.4 Recommended Implementation

When the ASD curve is nearly ‘V’-shaped over the range of delays that matters to the estimator, the built-in Brownian bias model may be sufficient. However, when the ASD curve has a clear mid-delay flat region and the PIM duration extends near to that flat region, the first-order Gauss-Markov approximation developed in Sections 2 and 3 is a better match to the desired time-correlation structure. In that case, the recommended implementation is to use `ImuFactor` or `ImuFactor2` for the preintegrated measurements and to model the bias evolution outside as discussed above.

This recommendation is not a claim that `CombinedImuFactor` is incorrect. Rather, `CombinedImuFactor` is appropriate for a Brownian bias model, whereas the present paper is targeting sensors whose Allan deviation suggests a non-zero correlation time and a finite steady-state bias covariance. A separate Gauss-Markov factor lets the estimator preserve GTSAM’s efficient preintegration machinery while matching the stochastic model that was fitted from the ASD curve. GTSAM also exposes an unstable `GaussMarkov1stOrderFactor` class [8]; whether that class is used directly or replicated in custom code, its parameters should be selected from the discrete-time equivalent of the chosen continuous-time model.

5.5 Integration Duration

A PIM assumes that one bias linearization point is adequate over its integration interval. Therefore the keyframe interval should remain well below the delay at which the ASD curve leaves the measurement-noise-dominated region and enters the

bias-dominated region. For the example of Fig. 2, this transition occurs at approximately $\tau = 10$ seconds. In that case, a PIM duration on the order of 1–3 seconds is a reasonable upper bound. The exact choice depends on computation, vehicle dynamics, and acceptable linearization error, but the general rule is the same: keep the integration interval comfortably shorter than the bias-correlation times one is trying to model.

In summary, GTSAM can use the same S_n , S_ζ , and correlation-time information derived from the ASD curve, but the factorization must match the assumed bias dynamics. For sensors whose Allan deviation exhibits a clear bias-instability plateau, `ImuFactor` or `ImuFactor2` together with a separate first-order Gauss-Markov bias factor is the most faithful implementation of the model developed in this paper.

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A Analysis by Itô Calculus

Ordinary differential equations involving noise processes, such as the bias drift of eqn. (6), are stochastic differential equations (Langevin equations). Analysis of stochastic differential equations requires special attention, using methods such as Itô calculus.

A.1 Standard Wiener Process

For a standard Wiener process (i.e., Brownian motion) W_t , an "increment" is the change in the value of the process over a tiny interval of time dt :

$$W_{t+\Delta t} = W_t + dW_t. \quad (15)$$

The random increment dW_t has two by two main properties:

Normal Distribution - It is a random variable drawn from a Gaussian distribution with a mean of 0 and a variance equal to the time step Δt :

$$dW_t \sim \mathcal{N}(0, \Delta t).$$

Independence - The value of dW_t at this moment is completely independent of any previous values. It has no memory.

A.2 Some Important Stochastic Calculus Remarks

The most important insight is that the Gaussian 'white' noise, $\omega_b(t)$, in eqn. (6) would need to be recognized as the "derivative" of a Wiener process, which should raise an immediate red flag because the Wiener process is not differentiable almost everywhere [9].

The proper way to write this kind of Stochastic Differential Equation (SDE) is in the *differential* form. For example, a first-order equation such as eqn. (6) becomes

$$dX(t) = F X(t) dt + B dW(t), \quad (16)$$

where $dW(t)$ is the increment of a Wiener process with $E[(dW(t))^2] = S^2 dt$. The symbol S^2 is the power spectral density (PSD) of the noise and S is the root PSD.

Now we are able to do the simulation of such an SDE by

$$X_{i+1} = X_i + F X_i \Delta t_i + B S \sqrt{\Delta t_i} Z_i, \quad (17)$$

where Z_i is a standard Gaussian random number (i.e., $Z_i \sim \mathcal{N}(0, 1)$) and $\Delta t_i = t_{i+1} - t_i$. This is called the Euler-Maruyama method [10].

Remark A.1: Beware of the $\sqrt{\Delta t_i}$ normalization factor! This is necessary by the definition of random increments discussed in Appendix A.1. $\triangle$

Remark A.2: The Euler-Maruyama method of eqn. (17) is an explicit discretization method. There exist implicit versions of these methods as well [10]. $\triangle$

A.3 Itô's Lemma

Stochastic processes can be analyzed more rigorously using Itô calculus.

Lemma A.3 (Itô's Lemma). *Let X_t be a diffusion process defined by the stochastic differential equation*

$$dX_t = a(X_t, t) dt + \sqrt{v(X_t, t)} dW_t,$$

where $a(x, t)$ is the infinitesimal mean and $v(x, t)$ is the infinitesimal variance. If $u(x, t)$ has continuous first-order derivatives in t and continuous first- and second-order derivatives in x , then $Y_t = u(X_t, t)$ is also a stochastic process that satisfies

$$du(X_t, t) = \left(\partial_t u + a \partial_x u + \frac{1}{2} v \partial_{xx} u \right) dt + \sqrt{v} \partial_x u dW_t. \quad (18)$$

Equivalently, in terms of dX_t :

$$du(X_t, t) = \partial_t u(X_t, t) dt + \partial_x u(X_t, t) dX_t + \frac{1}{2} \partial_{xx} u(X_t, t) v(X_t, t) dt.$$

The first two terms are the ordinary chain-rule terms that would appear if X_t were differentiable in time. The last term is the "Itô term," arising because the quadratic variation of a diffusion process satisfies $(dX_t)^2 = v(X_t, t) dt$ in the mean-square sense.

The Itô isometry for the standard Brownian motion dW_t and the stochastic process X_t is:

$$E \left[\left(\int_0^T X_t dW_t \right)^2 \right] = E \left[\int_0^T X_t^2 dt \right]. \quad (19)$$

It is useful for computation of variances of stochastic processes, as the following sections will demonstrate.

A.4 Analysis of Integrated White Noise

The integrated phase

$$\alpha(t) \triangleq \int_0^t z_N(s) ds \quad (\text{or } \dot{\alpha}(t) = z_N(t))$$

models the accumulated measurement noise from integrating gyro or accelerometer data to compute angle or velocity. The quantity $z_N(t)$ represents white measurement noise with PSD $S_n = N^2$ (corresponding to $n(t)$ in eqn. (5) and the ARW parameter N from Section 1.2). Equivalently, $z_N(t) = N \omega(t)$ where $\omega(t)$ is unit white noise, i.e., the formal derivative of a standard Wiener process.

The Itô differential form is

$$d\alpha(t) = N dW_t, \quad dW_t \sim \mathcal{N}(0, \Delta t), \quad (20)$$

where $d\alpha(t)$ is the infinitesimal phase increment of $\alpha(t)$ over one sample period Δt . Integrating both sides from 0 to t yields

$$\int_0^t d\alpha(s) = N \int_0^t dW_s \quad (21)$$

$$\alpha(t) - \alpha(0) = N \int_0^t dW_s \quad (22)$$

$$\alpha(t) = \alpha(0) + N \int_0^t dW_s. \quad (23)$$

Therefore, $E[\alpha(t)] = \alpha(0)$. The variance of $\alpha(t)$ is

$$\text{Var}[\alpha(t)] = E[(\alpha(t) - E[\alpha(t)])^2] \quad (24)$$

$$= E \left[\left(N \int_0^t dW_s \right)^2 \right] = N^2 E \left[\left(\int_0^t 1 dW_s \right)^2 \right].$$

Using the Itô isometry of eqn. (19) that

$$E \left[\left(\int_0^t 1 dW_s \right)^2 \right] = E \left[\int_0^t 1^2 ds \right]$$

yields

$$\text{Var}[\alpha(t)] = N^2 E \left[\left(\int_0^t 1^2 dt \right) \right] = N^2 t.$$

Remark A.4: Note that the units work out. In the case of a gyro, where $S_n = N^2$ has units of $\frac{(\text{rad/s})^2}{\text{Hz}} = \frac{\text{rad}^2}{s}$, then $\text{Var}[\alpha(t)] = N^2 t$ correctly has units of rad^2 . The ARW parameter N has units of $\text{rad}/\sqrt{s} = (\text{rad/s})/\sqrt{\text{Hz}}$. $\triangle$

Remark A.5: A common unit used for N on gyro data sheets is $\text{mdps}/\sqrt{\text{Hz}}$. This can be converted as

$$\begin{aligned} \text{mdps}/\sqrt{\text{Hz}} &= 10^{-3} \text{ deg}/s/\sqrt{\text{Hz}} \\ &= (\pi/180) 10^{-3} \text{ rad}/s/\sqrt{\text{Hz}} \\ &= 1.75 \times 10^{-5} \text{ rad}/s/\sqrt{\text{Hz}}. \end{aligned}$$

$\triangle$

Remark A.6: For the accelerometer, the manufacturer sometimes uses N with the unit, $\mu g/\sqrt{\text{Hz}}$, which can be converted as:

$$\begin{aligned} \mu g/\sqrt{\text{Hz}} &= (9.81)(10^{-6}) (m/s^2) / (1/s^{1/2}) \\ &= 9.81 \times 10^{-6} m/s^{\frac{3}{2}}. \end{aligned}$$

$\triangle$

A.5 Analysis of Gauss-Markov Model

The Ornstein-Uhlenbeck process [11], given by

$$\dot{x}(t) = -\mu x(t) + \omega(t), \quad (25)$$

is equivalent to the bias model in eqn. (6). Assume that $\mu > 0$ and $\omega(t)$ has unit PSD. The Itô differential form is

$$dx(t) = -\mu x(t) dt + dW_t. \quad (26)$$

Define $u(x, t) = e^{\mu t} x$, so $y(t) = u(x(t), t) = e^{\mu t} x(t)$. Applying Itô's Lemma, which is included in Appendix A.3, gives

$$\begin{aligned} dy(t) &= \partial_t u(x, t) dt + \partial_x u(x, t) dx(t) + \frac{1}{2} \partial_{xx} u(x, t) (dx(t))^2 \\ &= \mu e^{\mu t} x(t) dt + e^{\mu t} dx(t), \end{aligned} \quad (27)$$

because $\partial_{xx} u(x, t) = 0$. Substituting $dx(t) = -\mu x(t) dt + dW_t$ yields

$$\begin{aligned} dy(t) &= \mu e^{\mu t} x(t) dt + e^{\mu t} (-\mu x(t) dt + dW_t) \\ &= e^{\mu t} dW_t. \end{aligned} \quad (28) \quad (29)$$

Integrating from 0 to t gives

$$e^{\mu t} x(t) - x(0) = \int_0^t e^{\mu s} dW_s, \quad (30)$$

and therefore

$$x(t) = e^{-\mu t} x(0) + e^{-\mu t} \int_0^t e^{\mu s} dW_s \quad (31)$$

Taking expectation yields

$$E[x(t)] = e^{-\mu t} x(0). \quad (32)$$

Computing the variance of $x(t)$ requires more care. Combining eqns. (31) and (32):

$$\text{Var}[x(t)] = e^{-2\mu t} \text{Var} \left[\int_0^t e^{\mu s} dW_s \right]. \quad (33)$$

Using the Itô isometry stated in eqn. (19) yields:

$$\begin{aligned} \text{Var} \left[\int_0^t e^{\mu s} dW_s \right] &= E \left[\left(\int_0^t e^{\mu s} dW_s \right)^2 \right] - E \left[\int_0^t e^{\mu s} dW_s \right]^2 \\ &= E \left[\int_0^t (e^{\mu s})^2 ds \right] - E \left[\int_0^t e^{\mu s} dW_s \right]^2 \\ &= E \left[\int_0^t e^{2\mu s} ds \right] \\ &= \frac{1}{2\mu} (e^{2\mu t} - 1). \end{aligned} \quad (34)$$

Consequently, from eqn. (33),

$$\text{Var}[x(t)] = \frac{1}{2\mu} (1 - e^{-2\mu t}). \quad (35)$$

The above discussion assumed unit PSD for $\omega(t)$ such that dW_t was the increment of a standard Wiener process with $dW_t \sim \mathcal{N}(0, \Delta t)$. When $\omega(t)$ is continuous-time white noise with PSD, S_w , the noise increment scales to $dW_t \sim \mathcal{N}(0, S_w \Delta t)$. Consequently, the general result is

$$\text{Var}[x(t)] = \frac{S_w}{2\mu} (1 - e^{-2\mu t}). \quad (36)$$

This has a finite steady-state variance

$$\lim_{t \rightarrow \infty} \text{Var}[x(t)] = \frac{S_w}{2\mu}, \quad (37)$$

which is equivalent to eqn. (8). The units of S_w are discussed in Section B.

B Unit Analysis: Check

The equation

$$\sigma_\zeta^2 = \frac{1 - \phi^2}{2\mu} S_\zeta \quad (38)$$

from Section 2.3 is derived by setting eqn. (8) equal to eqn. (12).

To get comfortable with this equation and with the (sometimes confusing) units associated with PSD's, it is useful to perform an analysis of the units of each term in this equation.

Gyro: The units of the bias are rad/s . The covariance P has units $(\text{rad}/s)^2$. Therefore, from eqn. (11), the units of σ_ζ^2 are also $(\text{rad}/s)^2$, because all summed items have the same units. The time derivative of the covariance $\dot{P}$ has units $\frac{(\text{rad}/s)^2}{s}$. Therefore, each term being added in eqn. (7) including the PSD S_ζ has these same units. The numerator of eqn. (38) (i.e., $(1 - \phi^2)$) has no units. The denominator (2μ) has units

of reciprocal seconds. The units of the product on the right-hand side are thus the units of S_ζ (i.e., $\frac{(\text{rad}/s)^2}{s}$) divided by the units of μ (i.e., $\frac{1}{s}$), which yields $(\text{rad}/s)^2$.

Note that the driving noise ζ has units of $\frac{\text{rad}}{s^2}$. The units of its PSD are:

$$\frac{(\text{rad}/s)^2}{s} = \frac{\text{rad}^2}{s^3} = \frac{\text{rad}^2}{s^3} \frac{s}{s} = \frac{(\text{rad}/s^2)^2}{\text{Hz}}. \quad (39)$$

Accelerometer: The units of the bias are m/s^2 . The covariance P has units $(m/s^2)^2$. Therefore, from eqn. (11), the units of σ_ζ^2 are also $(m/s^2)^2$, because all summed items have the same units. The time derivative of the covariance $\dot{P}$ has units $\frac{(m/s^2)^2}{s}$. Therefore, each term being added in eqn. (7) including the PSD S_ζ has these same units. The numerator of eqn. (38) (i.e., $(1 - \phi^2)$) has not units. The denominator (2μ) has units of reciprocal seconds. The units of the product on the right-hand side are thus the units of S_ζ (i.e., $\frac{(m/s^2)^2}{s}$) divided by the units of μ (i.e., $\frac{1}{s}$), which yields $(m/s^2)^2$.

Note that the driving noise ζ has units of $\frac{m}{s^3}$. The units of its PSD are:

$$\frac{(m/s^2)^2}{s} = \frac{m^2}{s^5} = \frac{m^2}{s^5} \frac{s}{s} = \frac{(m/s^3)^2}{\text{Hz}}. \quad (40)$$

In both cases, the unit analysis shows that both the left and right sides of eqn. (38) have the same units (as they should) and that the units of the PSD are the units of the driving noise squared per Hz.

C Approximate Discrete-Time Driving Noise Covariance

Due to the fact that $\phi = \exp(-\mu T_s)$,

$$\begin{aligned} 1 - \phi^2 &= 1 - (1 + (-2\mu T_s) + 0.5(-2\mu T_s)^2 + \dots) \\ &= 2\mu T_s - 0.5(2\mu T_s)^2 + \dots \end{aligned}$$

When $\mu T_s \ll 1$, which is typically the case by design, because the sample period T_s is much less than the correlation time of the IMU bias, the higher-order terms are small and may be dropped:

$$1 - \phi^2 \approx 2\mu T_s.$$

In this case, eqn. (38) simplifies to

$$\sigma_\zeta^2 \approx \frac{2\mu T_s}{2\mu} S_\zeta = T_s S_\zeta. \quad (41)$$